

SIMATS ENGINEERING

CO3_AT1_Assessment Tool: EXPERIMENT

DSA06 – Data Handling and Visualization | Assessment Tool: Experiment

DSA06 – DATA HANDLING AND VISUALIZATION

ASSESSMENT TOOL I – EXPERIMENTS

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No: 192324189 Name: N HARSHAVARDHAN

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool: Experiments

Weightage: 20%

Code of Conduct:

I N HARSHAVARDHAN (192324189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N HARSHAVARDHAN

Criteria	Understanding of Concepts 25%	Experimental Design 30%	Use of Tools 20%	Analysis of Results 15%	Documents 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

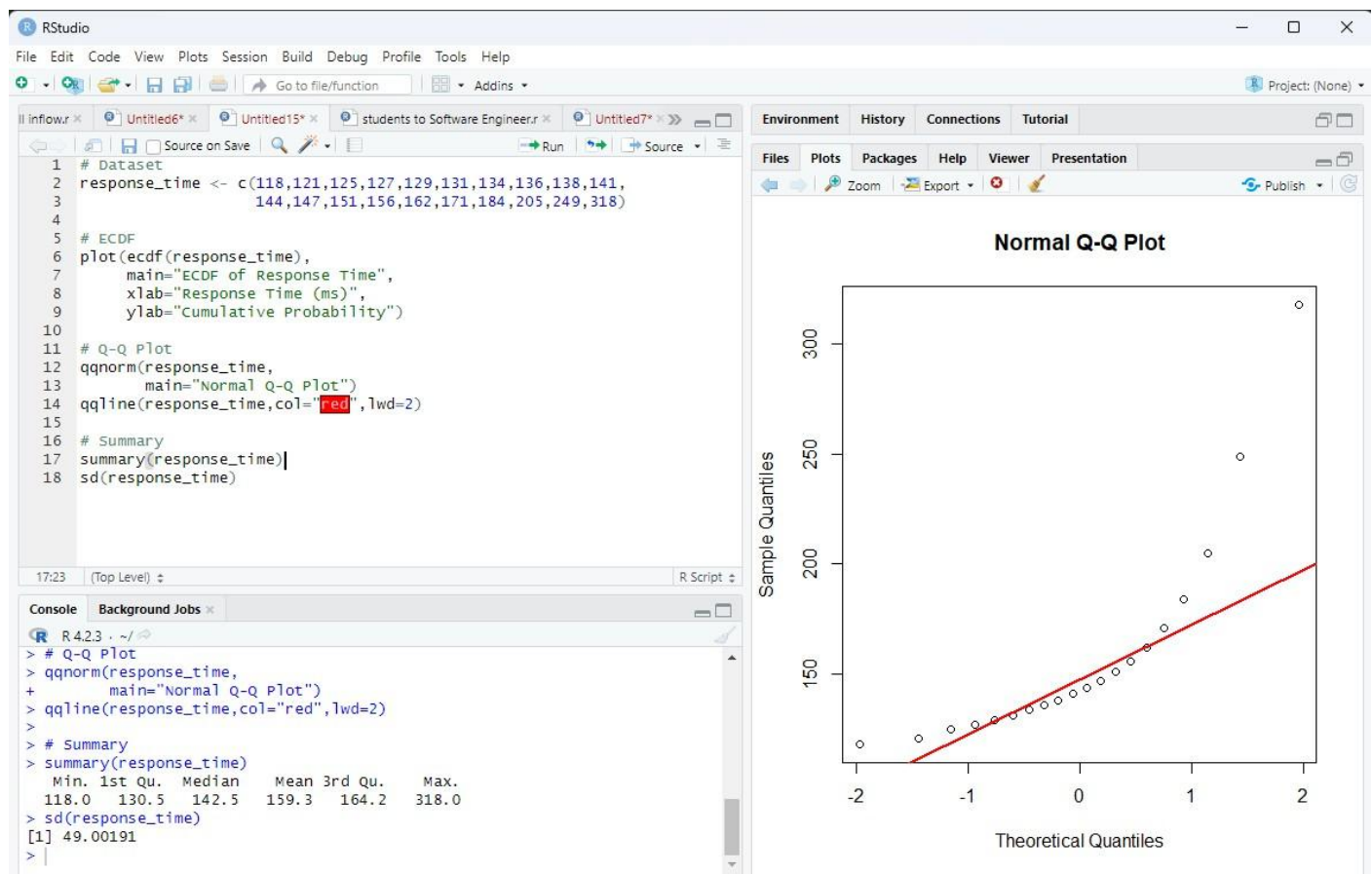
Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

Tasks:

1. Construct an ECDF and a normal Q-Q plot.
2. Use both plots to evaluate the normality claim.
3. Identify evidence of skewness and/or extreme observations.
4. Recommend percentile-based limits or mean $\pm$ standard-deviation limits, with justification.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation



Question 1

```
response_time <- c(118,121,125,127,129,131,134,136,138,141,
                  144,147,151,156,162,171,184,205,249,318)
```

ECDF

```
plot(ecdf(response_time),  
     main = "ECDF of Response Time",  
     xlab = "Response Time (ms)",  
     ylab = "Cumulative Probability")  
  
# Normal Q-Q Plot  
  
qqnorm(response_time,  
        main = "Normal Q-Q Plot")  
  
qqline(response_time)  
  
# Summary statistics  
  
mean(response_time)  
  
sd(response_time)  
  
quantile(response_time, c(0.05,0.95))
```

Expected Output

The program produces:

1. An ECDF plot showing the cumulative distribution.
2. A Normal Q-Q plot with most lower observations near the line but the largest observations deviating upward.
3. Mean, standard deviation, and 5th/95th percentiles.

Analysis

- The ECDF increases gradually for most observations but rises slowly at the upper end because of very large values.
- The Q-Q plot shows noticeable upward deviation for 205, 249, and 318 ms.
- This indicates positive (right) skewness and the presence of extreme observations.
- Therefore, the data is not approximately normal.

Recommendation

Percentile-based limits are preferable to mean $\pm$ standard deviation limits because the distribution is skewed and contains extreme values. Percentiles are less affected by outliers.

Conclusion

The normality claim is not supported. The response-time data is right-skewed with extreme high values, so percentile-based monitoring limits are recommended.

Question 2 – Comparing Many Distributions and Bean Plots

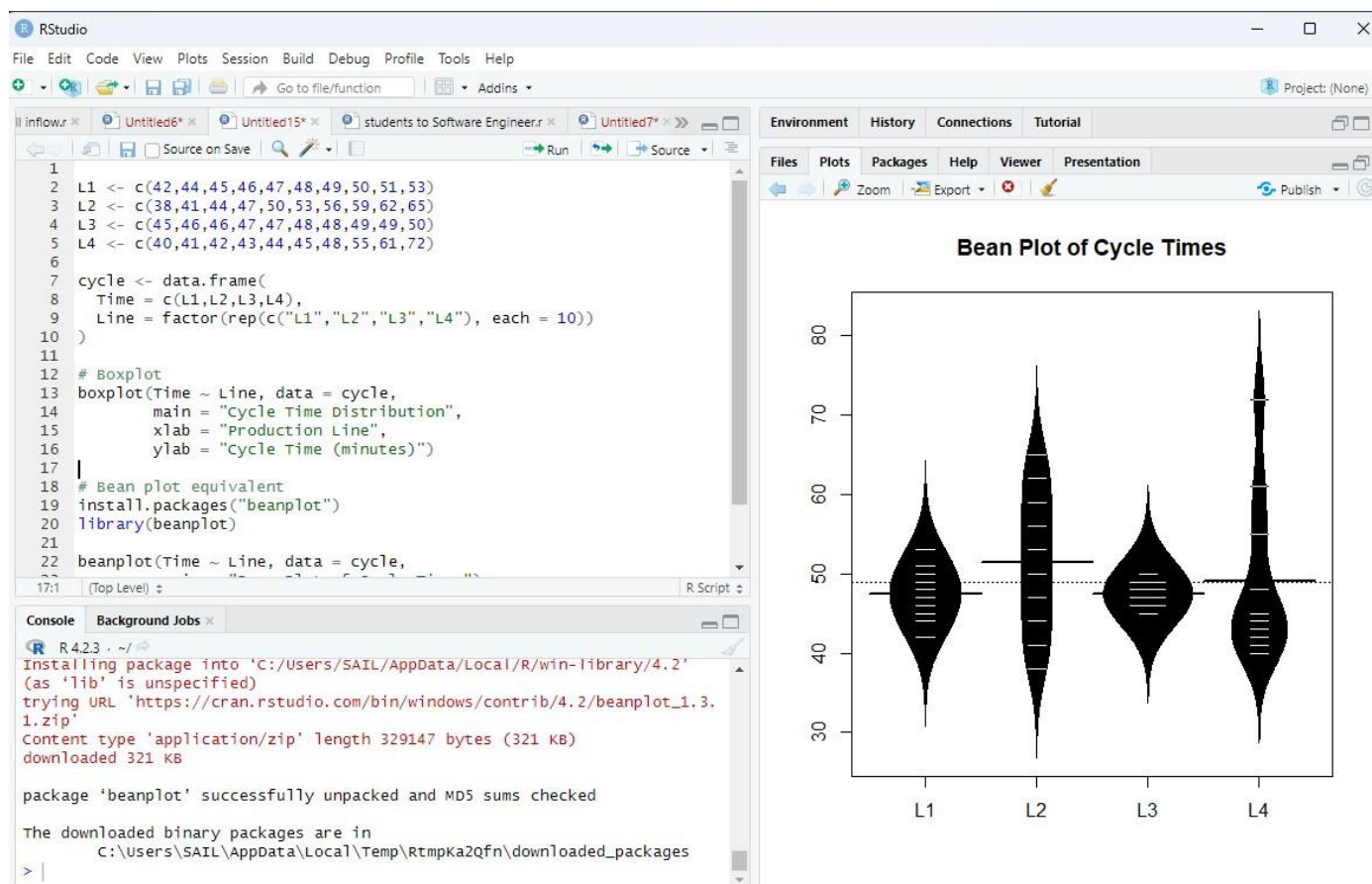
Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Production Line	Cycle Times (minutes)
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. Create a visualization comparing all four distributions.
2. Construct bean plots or an equivalent distribution visualization along the vertical axis.
3. Determine which line has greatest variability and which has the most concentrated distribution.
4. Explain why comparing only means is insufficient.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability



Question 2

```
L1 <- c(42,44,45,46,47,48,49,50,51,53)
```

```

L2 <- c(38,41,44,47,50,53,56,59,62,65)

L3 <- c(45,46,46,47,47,48,48,49,49,50)

L4 <- c(40,41,42,43,44,45,48,55,61,72)

cycle <- data.frame(

  Time = c(L1,L2,L3,L4),

  Line = factor(rep(c("L1","L2","L3","L4"), each = 10))

)

# Boxplot

boxplot(Time ~ Line, data = cycle,

  main = "Cycle Time Distribution",

  xlab = "Production Line",

  ylab = "Cycle Time (minutes)")

# Bean plot equivalent

install.packages("beanplot")

library(beanplot)

beanplot(Time ~ Line, data = cycle,

  main = "Bean Plot of Cycle Times")

# Standard deviations

tapply(cycle$Time, cycle$Line, sd)

```

Expected Output

The output contains:

- A boxplot comparing all four lines.
- A bean plot showing distribution shape and density.
- Standard deviations for each production line.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

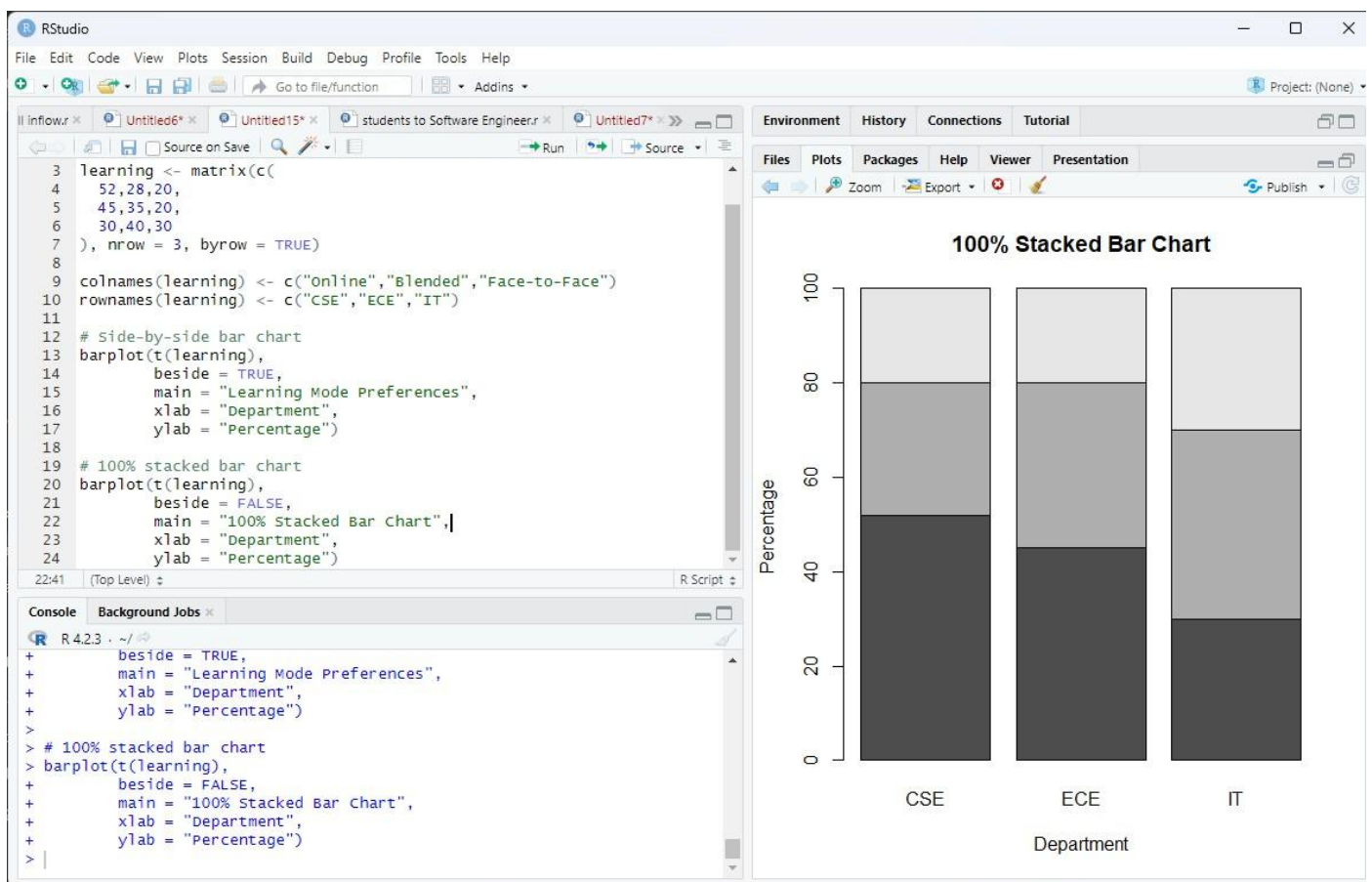
Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.

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2. Explain which chart better supports comparison of individual proportions and why.
3. Explain why separate pie charts may be less effective for comparing departments.
4. Identify one design choice that could make small proportional differences appear larger.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions



Expected Output

Two charts are generated:

1. Side-by-side grouped bar chart.
2. 100% stacked bar chart.

Analysis

- The side-by-side bar chart is better for comparing individual proportions because bars for the same learning mode share a common baseline.
- The stacked chart is useful for showing the composition of each department.
- Separate pie charts are less effective because comparing angles across multiple pies is difficult.
- Starting the y-axis above zero or using a truncated axis can exaggerate small differences.

Conclusion

For comparing proportions across departments, the side-by-side bar chart is the clearer and more accurate choice.

Question 4 – Nested Proportions using Treemap and Sunburst

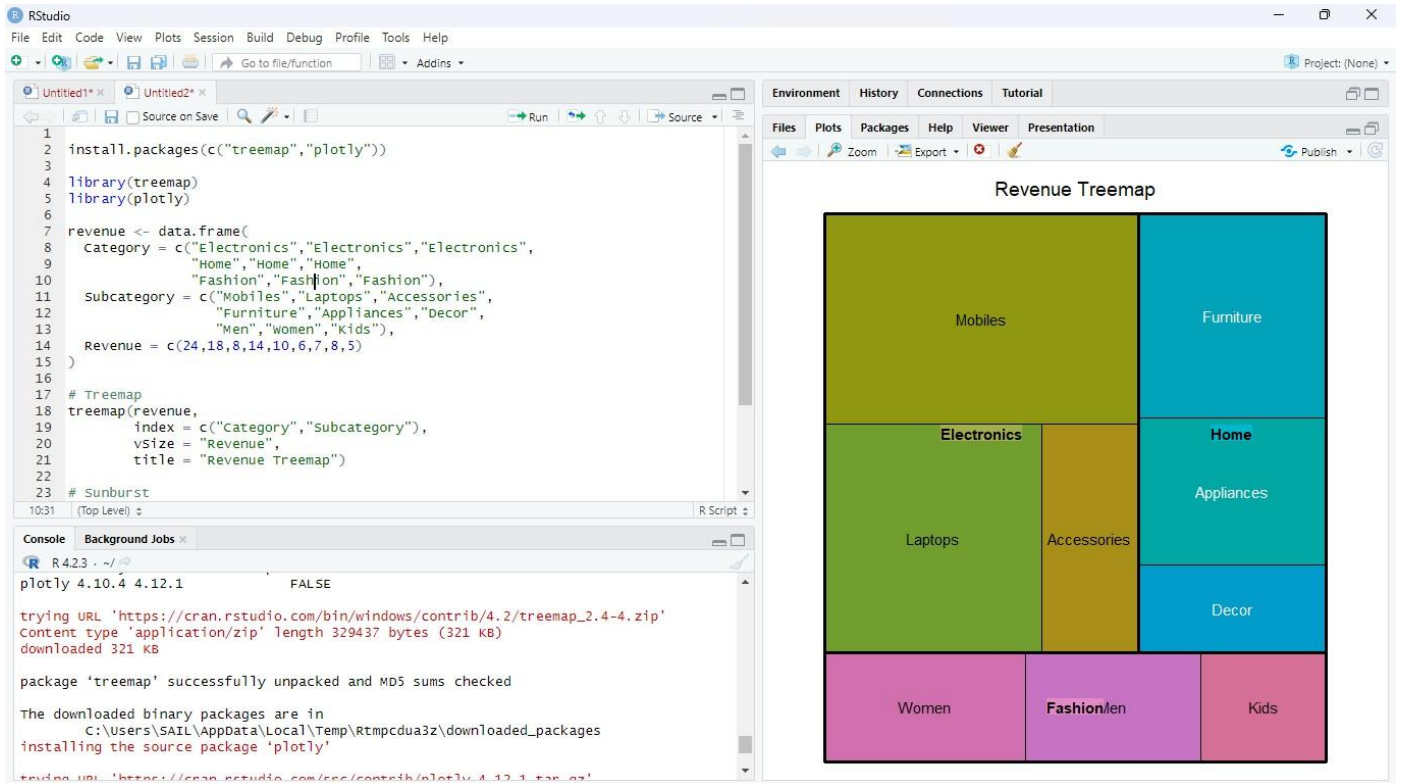
Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:

1. Construct a treemap showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.
3. Determine the largest category and calculate its percentage of total revenue.
4. Compare treemap and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

Key competency: Treemaps, sunburst charts, nested proportions, area encoding



Expected Output

- A treemap where rectangle area represents revenue.
- A sunburst chart showing category and subcategory hierarchy.

Analysis

Category totals:

Category	Revenue (₹ lakh)
Electronics	50
Home	30
Fashion	20

Conclusion

Electronics is the largest category, contributing 50% of the listed total revenue. Treemaps emphasize size, while sunbursts emphasize hierarchical structure.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

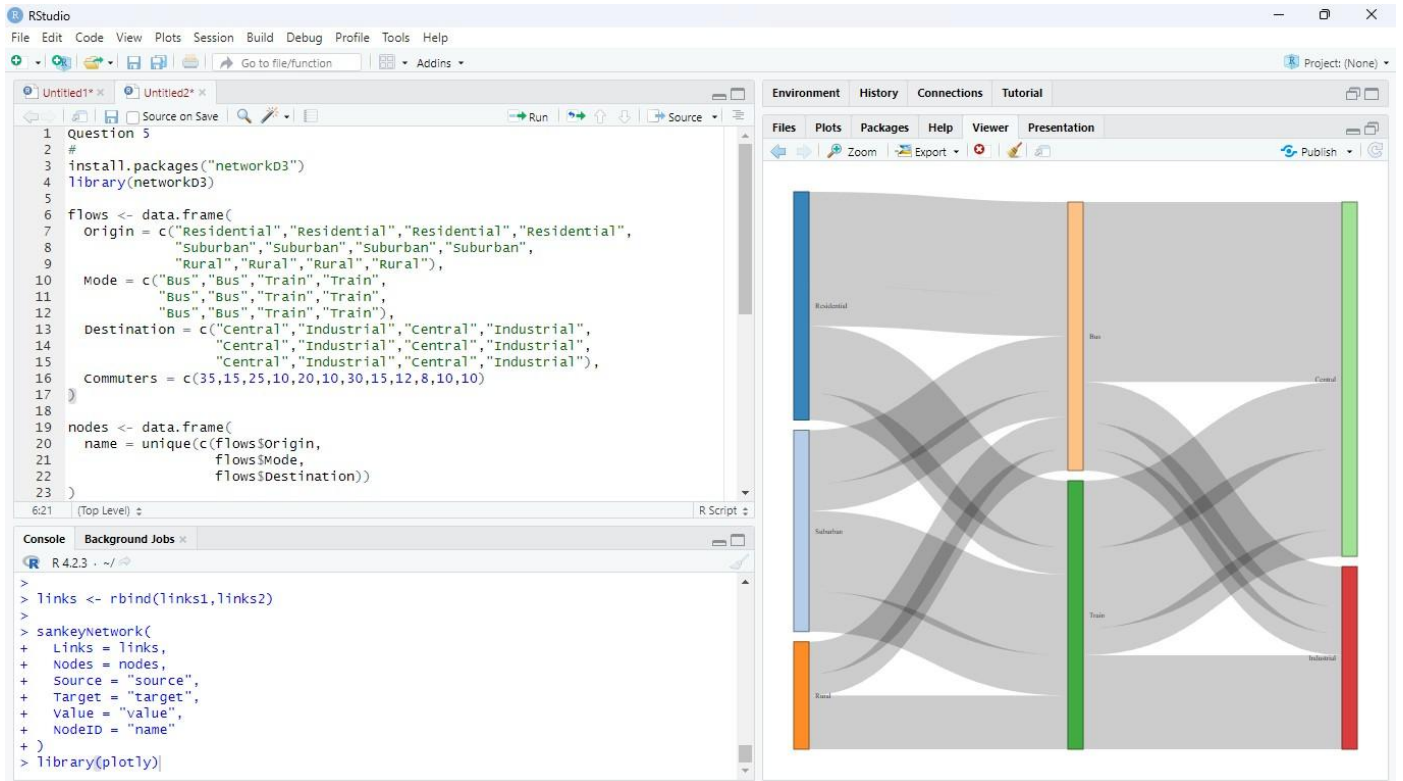
Tasks:

1. Construct a Sankey diagram showing Origin → Mode → Destination.
2. Construct a parallel-sets representation of the same categorical flows.

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3. Identify the three largest flows and calculate their percentages of all commuters.
4. Compare Sankey and parallel sets for flow magnitude and category relationships.
5. Explain one way poor link widths or category ordering could mislead the viewer.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships



Suggested Evaluation Rubric – Experiment

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.